Lesson notes · for you

Defenders: cryptography, networks, careers: unit check

Cyber security · Level 4 · Unit check

Time	Materials
10 min	Paper and a pen. No device needed.

Learning goals

- Fingerprints of data: hashing
- Locks and keys: public-key cryptography
- Networks and the defenders who watch them
- Careers in cyber security

Activity: Review: Defenders' quiz

1. Recall (3 min): Without notes, write one idea you remember from each of the four lessons.
2. Quiz (5 min): Take turns asking a partner one question per lesson. The partner answers and explains why.
3. Fix (2 min): Write one habit you will change, such as a unique passphrase for your email, or never "just looking" where you have no permission.

Work in pairs. Keep this short: the unit check comes next.

Discussion

- Why can't a good website send you your password?
- What does the padlock prove, and what doesn't it?
- What would you do if you found a weakness by accident?

Check yourself

- The idea that surprised me most, and one step I will take.

Try it this week
Pick the idea from this unit that surprised you most, explain it to someone at home, and write down one step you will take: to keep your own accounts safer, or towards a career in security.

Worksheet

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Name _____ Class _____ Date _____

1 **You hash a single word and then a whole film with SHA-256. How long is each fingerprint?**

- ◯ Both are 256 bits: 64 hexadecimal characters
- ◯ The film's is much longer than the word's
- ◯ As long as the input, but scrambled

2 **A website that stores only salted hashes can email you your old password if you forget it.**

- ◯ True
- ◯ False

3 **Which statements about public and private keys are true?**

- ☐ Anyone can lock a message for you with your public key
- ☐ Only your private key opens a message locked with your public key
- ☐ Send your private key to friends so they can write to you
- ☐ Anyone can check your signature with your public key
- ☐ The HTTPS padlock proves a site is honest

4 **Put the steps of signing in to a well-built website in order.**

- ☐ If they match, you're signed in
- ☐ It compares the result with the stored fingerprint
- ☐ The site adds your salt and hashes what you typed
- ☐ You type your password

5 **Someone quietly changes the fees list on the university finance office's computer. Which security goal is broken?**

- ◯ Confidentiality
- ◯ Integrity
- ◯ Availability